

Department of Mathematical and Computational Sciences
National Institute of Technology Karnataka, Surathkal

Assignment 3 – MA725

Submission Date: _____

Instructions:

- Answer all questions.
 - Show all necessary calculations clearly.
 - Write neatly and clearly.
 - **Do not copy from each other. Plagiarism will result in zero marks.**
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1. Image inpainting is the process of reconstructing missing or corrupted regions of an image using surrounding pixel information. A commonly used mathematical model for inpainting is based on solving the two-dimensional Laplace equation inside the damaged region while keeping known pixels fixed as boundary conditions. Consider a grayscale image or feature matrix provided in a file named `matrix.txt`¹, where the matrix entries represent pixel intensities and missing pixels are indicated by zero values. Starting from the continuous Laplace equation, derive the finite-difference discretization using the five-point stencil and explain how the resulting linear system can be solved using the SOR iterative method. Describe how a binary mask is used to identify damaged pixels and restrict updates during iteration. Further, develop an algorithm and implement a Python program to perform image inpainting using the SOR method, and demonstrate the reconstruction of the damaged image by displaying the original matrix, the mask, and the inpainted image. Discuss the convergence behavior of the method and comment on the quality of reconstruction obtained.
2. Principal Component Analysis (PCA) can be employed as an anomaly detection technique by analyzing the reconstruction error obtained after projecting data onto a low-dimensional subspace. Using the Olivetti faces dataset available through `sklearn.datasets.fetch-olivetti-faces`, load and pre-process the face images, and artificially introduce anomalies by flipping one or more images upside down. Apply PCA to encode and reconstruct both normal and anomalous images, and compute the corresponding reconstruction errors to demonstrate that anomalous samples yield significantly higher errors. Visualize the original images, reconstructed images, and reconstruction error maps. Subsequently, repeat the same experiment using Singular Value Decomposition (SVD) directly, without relying on the PCA implementation in `sklearn`, by computing the average face, centering the data matrix, and applying `np.linalg.svd` to the transposed dataset. Use the leading singular vectors to perform image encoding and decoding, and explain why the decoding step represents the inverse transformation. Finally, discuss the effect of the number of retained components on reconstruction quality, justify why $U_i U_i^T$ approaches the identity matrix as the number of components increases, and comment on the relationship between PCA and data-driven basis representations analogous to the Fourier transform.

¹https://courses.iris.nitk.ac.in/pluginfile.php/702754/mod_resource/content/1/image.txt